

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
8th Semester of B. Pharm. Examination
University Theory Examination May 2017
PH413 NOVEL DRUG DELIVERY SYSTEM

Date: 12/05/2017, Friday

Time: 1:30 p.m to 4:30 p.m

Maximum Marks: 80

Instructions:

- 1. Figures to right indicate marks.**
- 2. Section wise separate answer book to be used.**
- 3. Draw neat sketches wherever necessary.**

SECTION- I

Q 1 Attempt any **TWO** of the following

A

1. Explain the routes of drug absorption through skin and suitable drug candidate for TDDS. **05**
2. Enlist various permeation enhancement techniques for Transdermal Drug Delivery System and describe any one in detail. **05**

B

1. Describe gradient controlled Transdermal Drug Delivery System. **05**
2. Write down evaluation parameters for niosomes. **05**

C Evaluation of Transdermal Drug Delivery System . **10**

Q 2 Attempt any **TWO** of the following

A Draw and label the Structure of : **10**
1. Niosomes 2. Liposomes and 3. Nanoparticles (Draw both type of structure)

B

1. Give classification of liposomes and methods of drug loading in liposomes. **05**
2. Describe approaches of drug targeting. **05**

C

1. Write down advantages of drug targeting. **04**
2. Describe levels of drug targeting. **06**

SECTION- II

- Q 3** Attempt any TWO of the following
- A** Define terms like sustained release, controlled release and extended release. Give Merits, Demerits and rational of designing modified release dosage form. **10**
 - B** Classify Osmotic drug delivery systems. Write down factors influencing drug release from the Osmotic Tablet. **10**
 - C** Compare and differentiate oral immediate and matrix drug delivery system. Enlist polymers used in Matrix drug delivery systems. What are the different drug release mechanisms associated with matrix drug delivery system. **10**
- Q 4** Attempt any TWO of the following
- A** Classify controlled release dosage form and explain Factors influencing the design of controlled release dosage form **10**
 - B** Discuss in detail the components of osmotic drug delivery system. Discuss push pull type of osmotic pump. **10**
 - C** Write on different mechanisms by which gastroretention of dosage forms is attained. Discuss hydrogel based Gastro retentive systems. **10**
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